

Interim Project Report

CyberNova — Online Scam Detection and Reporting System

Prepared for Hume City Council CyberSafe Services

Group name: CyberKent

Executive summary

Group **CyberKent** is developing **CyberNova**, an Online Scam Detection and Reporting System for **Hume City Council CyberSafe Services**. The platform lets residents, small businesses, community organisations and not-for-profits check suspicious messages, links, phone numbers and email addresses; report scams formally with evidence; follow moderated community alerts and a regional scam map; and work through guided recovery steps after being targeted. Council staff use the same platform to review, verify, de-duplicate and publish, and to produce statistics and audit trails.

At the interim point the analysis and design phases are complete and verified against one another. The requirements baseline holds **72 functional requirements across 12 modules, 30 non-functional requirements and 6 ethical requirements**. The technology stack, the modular-monolith architecture and the UI design system are all baselined. A working public landing page and an interactive scam-check prototype have been built and are running, which demonstrates the front-end stack end to end. The backend workspace is scaffolded and database schema design begins next.

The feasibility study returns a **GO recommendation, subject to five conditions** (Section 8.7). The project is technically feasible on a wholly free and open-source stack; economically compelling, with an indicative benefit–cost ratio between **3.5:1 and 13:1** depending on how effective the service proves; operationally feasible provided Council commits approximately 0.4 FTE of reviewer capacity; and legally feasible under the Privacy Act 1988 (Cth), the Australian Privacy Principles and the Victorian Privacy and Data Protection Act 2014, provided the data-handling controls in Section 11 are implemented as designed.

Cash cost to build is **AUD \$0** — the project runs entirely on free tiers with student labour, a notional value of approximately **AUD \$64,800**. Indicative annual operating cost once promoted publicly is approximately **AUD \$4,400** for infrastructure, rising to roughly **AUD \$28,400** including part-time maintenance. Six funding avenues are identified in Section 7.3.

The single most important finding of the interim phase is set out in Section 11. Because this system stores accounts of financial and emotional harm, contact indicators that may belong to innocent people whose identities were spoofed, and evidence files that frequently contain more personal data than the reporter realises, **the ethical risk in this project concentrates in the database design rather than the user interface**. Fifteen distinct ethical risks in the data model have been identified and each has been assigned concrete, testable mitigation actions, most of which must be built into the schema before the first migration is written. That work is the immediate next milestone.

PROJECT CHARTER

Field	Detail
Group name	CyberKent
Project name	CyberNova — Online Scam Detection and Reporting System
Client	Hume City Council, Victoria, Australia
Business unit	Hume City Council CyberSafe Services
Document version	1.0
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Stakeholders:				

Revision history

Version	Date	Description	Author
0.1	30 Jun 2026	Project initiation, client brief analysed	CyberKent
0.2	8 Jul 2026	Requirements baseline drafted (FR1–FR72, NFR-1–NFR-30)	CyberKent
0.5	15 Jul 2026	Technology stack, architecture and UI design system baselined	CyberKent
0.9	16 Jul 2026	Interim SRS Report v1.0 issued	CyberKent
1.0	1 Aug 2026	Interim Project Report consolidated and issued for review	CyberKent

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2. Project background

2.1 The client

Hume City Council is a local government area roughly 20 kilometres north of central Melbourne, and one of Australia's fastest-growing municipalities. It serves **more than 262,000 residents** across established urban suburbs in the south and rural land in the north. Council headquarters are at **1079 Pascoe Vale Road, Broadmeadows**.

The municipality contains Melbourne Airport, the Hume and Calder Highways and the Western Ring Road, which makes it a transport, freight and employment hub for the northern region and the state.

Council delivers a broad service portfolio that gives this project its context:

- **Community services** — maternal and child health, immunisation, assertive youth outreach for ages 10 to 24, aged care and disability support.
- **Environment and sustainability** — management of more than 160,000 trees, the Hume Home Energy Help program, climate action planning and Green Wedge protection.
- **Advocacy and growth** — as a fast-growing interface council, Hume actively advocates to State and Federal government for investment in infrastructure, transport and **community safety**.

CyberSafe Services is the arm of that community safety work that addresses online harm. CyberNova is the digital service through which it will operate.

2.2 Why this project, and why now

Three characteristics of the Hume community make it disproportionately exposed to online scams.

A highly diverse, multilingual population. Scam messaging exploits unfamiliarity with how Australian institutions actually communicate. Residents who have arrived recently, or who read English as a second language, have fewer reference points for recognising a fake myGov, bank, Australia Post or Australian Taxation Office message.

A large base of small and micro businesses and volunteer-run organisations. Small businesses, community groups and not-for-profits in the municipality carry real financial exposure — invoice redirection, business email compromise, payment fraud — but almost none of them have in-house IT security, a security budget, or anyone whose job it is to think about this.

A demographic profile that includes many people at elevated risk. Older residents, young people aged 10 to 24 already served by Council's youth outreach, and people experiencing financial stress are all groups that scam operators target deliberately.

Against that exposure, the currently available help is fragmented. National services such as Scamwatch and the National Anti-Scam Centre operate at a national scale and cannot tell a Broadmeadows resident which scams are circulating in Broadmeadows this month. Bank and telecommunications warnings reach only their own customers. Council itself has no system that lets it see what is actually happening to its own community, which means it cannot target its awareness work, cannot answer a resident who asks "is this real?", and cannot evidence the problem when it advocates for funding.

2.3 The origin of the engagement

Council issued a project brief to the university development program requesting an online cybersecurity portal for small businesses, community organizations and not-for-profits. The brief specified 14 system modules, 72 functional requirements, 30 non-functional requirements and a set of ethical requirements, and required the development team to research and propose an additional 15 to 20 enhanced functional requirements for stakeholder approval.

Group CyberKent was formed to deliver it. The brief left three areas explicitly to the development team: the hardware recommendation, the software recommendation, and the enhanced functional requirements. It also contained internal inconsistencies which the team identified during analysis and has raised formally rather than resolving unilaterally (see Section 5.1.4 and the Interim SRS Report Section 11).

2.4 What already exists, and what this project adds

Existing channel	What it does	Gap this project fills
Scamwatch / National Anti-Scam Centre	National scam reporting and national trend reporting	No municipal-level visibility; no local alerting; no feedback loop to Council
Bank and telco warnings	Warn their own customers about scams affecting their own products	Reach only customers; siloed; nothing for community organisations
ACSC / Cyber.gov.au guidance	General cybersecurity advice for businesses	Generic; no "is this specific message a scam?" capability; not targeted to local context
Council newsletters and social media	Occasional awareness posts	Ad hoc, not evidence-driven, no reporting path, no analytics
CyberNova (this project)	Municipal checking, reporting, verification, alerting, education and recovery, with local analytics	Provides the local layer none of the above provides

CyberNova is deliberately **complementary, not competing**. It does not replace Scamwatch, IDCARE, the police or a bank's fraud line — it routes people to them at the right moment, and it gives Council the local evidence base it currently lacks.

3. Problem statement

3.1 Statement of the problem

Residents, small businesses, community organisations and not-for-profit organisations in the City of Hume are exposed to online scams that are increasing in volume and sophistication, but they have **no local, trusted, immediate way to check whether a suspicious message is a scam, no simple way to report one, and no reliable source of warnings about the scams currently circulating in their own community**. At the same time, Hume City Council has **no visibility of the scam activity affecting its residents**, so it cannot target its community safety work, cannot warn people early, cannot direct victims to recovery support, and cannot evidence the problem when it advocates for State and Federal investment. The result is preventable financial loss, avoidable psychological harm, under-reporting, duplicated Council effort, and a community safety function operating without data.

3.2 Decomposition of the problem

The statement above resolves into six specific, addressable problems.

#	Problem	Who it affects	Consequence today	Requirements that address it
P1	No way to verify a suspicious message before acting on it. A resident receiving an unexpected "your parcel is held" SMS has no trusted place to ask.	All community members	People either act on scams, or ignore genuine communications out of fear	FR13–FR24 (analysis modules)
P2	Reporting is difficult, fragmented and offers no feedback. Existing reporting paths are national, slow, and return nothing to the person reporting.	Victims and near-victims	Severe under-reporting; the true scale is invisible	FR25–FR30, FR61–FR63
P3	Warnings arrive too late, or not at all. By the time a scam is widely publicised, the local wave has passed.	Whole community	Repeat victimisation across the same suburb within days	FR49–FR54, FR64–FR65
P4	No local intelligence for Council. Council cannot see what is happening in its own municipality.	Council, CyberSafe Services	Awareness campaigns are generic and untargeted; advocacy lacks evidence	FR70–FR71, scam map/trends
P5	Victims do not know what to do next. After a scam, the sequence of actions (bank, passwords, IDCARE, police, credit ban) is not obvious and is time-critical.	Victims	Recoverable losses become unrecoverable; distress is prolonged	FR55–FR60
P6	Small organisations have no security capability at all. No IT staff, no budget, no assessment of their own exposure.	Small business, NFPs, community groups	Structural vulnerability; one incident can be existential	FR13–FR24, FR55–FR59

3.5 The proposed solution in one paragraph

CyberNova gives the Hume community a single trusted municipal portal that answers "is this a scam?" in seconds with a transparent, explained risk score; accepts a structured scam report with evidence and tells the reporter what happens next; lets trained Council staff verify reports and publish anonymised community alerts fast; shows regional scam activity on a map; provides categorised awareness material and step-by-step recovery checklists; and produces the de-identified statistics Council needs to target its work and evidence its advocacy — all while collecting the minimum personal data necessary, explaining every automated judgement it makes, and never presenting itself as professional cybersecurity certification.

4. General description of the project

4.1 Project details

4.1.1 Identification

Attribute	Detail
Project name	CyberNova
System name	Online Scam Detection and Reporting System
Group name	CyberKent
Client	Hume City Council, Victoria, Australia
Client business unit	Hume City Council CyberSafe Services
Project type	New-build, greenfield. No existing system is being replaced
Delivery model	Cloud-hosted responsive web application (Software as a Service)
Project duration	30 June 2026 – 27 November 2026 (22 weeks)
Current phase	Design complete; database design and implementation commencing
Methodology	Iterative and incremental, with fixed milestone gates and stakeholder review at each iteration

4.1.2 What the system does

CyberNova is a web-based cybersecurity portal delivering four capabilities to the community and four to Council.

For the community:

15. **Check** — submit suspicious message text, a URL, a phone number or an email address and receive a numeric risk score, a risk band (Low / Medium / High) and a plain-language explanation of every indicator that contributed to the score.

16. **Report** — lodge a structured scam report, attach evidence, receive a reference number, track the report's status and respond to requests for further information.
17. **Stay informed** — read moderated, anonymised community scam alerts; view regional scam activity and trends on a map; browse categorised awareness and education material; subscribe to alerts by scam category and by region.
18. **Recover** — follow guided step-by-step response checklists after being targeted, and record which recovery actions have been completed.

For Council:

19. **Review and verify** — work a prioritised review queue, assign reviewers, classify reports against a managed taxonomy, set severity, detect duplicates and related reports, request more information, and approve or reject with a recorded reason.
20. **Alert** — draft community alerts from verified reports, anonymise reporter information, obtain approval, publish, search and archive.
21. **Administer** — manage user accounts, roles and permissions, and maintain the scam category taxonomy.
22. **Analyse and audit** — produce statistics and dashboards on volumes, categories, severity, status and regional trends; export de-identified aggregate data; and read an immutable audit trail of security-relevant activity.

4.1.3 Technical summary

Layer	Technology	Reason
Frontend	React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui	Type-safe, component-based, accessible by default
Backend	Node.js, Express.js, REST over HTTPS	Single entry point, stateless, mobile- and AI-ready
Database	PostgreSQL with Prisma ORM	ACID compliance, strong concurrency, type-safe migrations
Authentication	JWT access and refresh tokens, bcrypt	Stateless auth; passwords never stored in plaintext
Security middleware	Helmet, express-rate-limit, Zod validation, TLS 1.3	Layered defence at every request
Files, maps, charts	Cloudinary, Leaflet with OpenStreetMap, Recharts	Evidence storage with CDN, scam map, dashboards
Email and logging	Nodemailer, Winston	Verification and notifications; structured audit logging
AI (future phase)	FastAPI (Python), OpenCV, EasyOCR, PyTorch	Isolated AI microservice; core system works without it
DevOps and hosting	Docker, Git/GitHub, Vercel, Render, Neon, UptimeRobot	Containerised, version-controlled, free-tier launch

The architecture is a **modular monolith with service-oriented boundaries**: one deployable backend in which each business capability owns its own controller, routes, validation, service, repository, DTOs and models, and communicates with other modules only through defined service interfaces. This keeps development simple now while allowing individual modules to be extracted into separate services later. Artificial intelligence sits behind an **AI Gateway** as a separate intelligence layer, so that if the AI platform is unavailable, core reporting continues to work.

Every core technology is free and open-source or has a usable free tier, as required by the project's zero-licensing-budget constraint.

4.2 Purpose of the project

The purpose of CyberNova is:

To reduce the financial and psychological harm caused by online scams in the City of Hume, by giving the community an immediate, trusted and accessible way to check, report and recover from scams, and by giving Hume City Council the local intelligence it needs to warn residents early, target its community safety work, and support victims — without creating fear, without collecting more personal data than the service requires, and without presenting automated results as professional cybersecurity certification.

Three commitments are embedded in that purpose and constrain every design decision that follows.

It must not create fear. People arrive at this service frightened, sometimes hours after losing money. A system that amplifies alarm to appear authoritative would cause harm. Tone, language, colour and result presentation are all governed by this.

It must be honest about what it is. The system produces advisory guidance based on the information the user supplies. It is not certification, it cannot guarantee protection, and serious incidents need qualified professionals or the relevant authority. This is stated on every result.

It must be explainable. Every risk score is shown with the indicators that produced it, in plain language. A score a person cannot interrogate is a score they cannot act on, and an automated judgement that cannot be challenged is one that cannot be corrected.

4.3 Objectives of the project

Objectives are expressed in SMART form and are traceable to the requirements baseline.

4.3.1 Product objectives

ID	Objective	Measure of success	Traceability
PO-1	Deliver a scam checking capability that analyses submitted text, URLs, phone numbers and email addresses and returns an explained risk score	All of FR13–FR24 implemented and passing acceptance tests; every score displays its contributing indicators	FR13–FR24, ETH-3

ID	Objective	Measure of success	Traceability
PO-2	Deliver an end-to-end scam reporting workflow from draft through submission, evidence attachment, reference number and status tracking to withdrawal	All of FR25–FR36 implemented; reporter can complete a report with evidence in under 10 minutes in usability testing	FR25–FR36
PO-3	Deliver a staff review and verification workflow including queue, assignment, classification, severity, duplicate detection and approval	All of FR37–FR48 implemented; every decision written to the audit log with a recorded reason	FR37–FR48, FR72
PO-4	Deliver moderated, anonymised community alerts with search, subscription and archiving	All of FR49–FR54 and FR64–FR66 implemented; no alert publishable while reporter-identifying data remains	FR49–FR54, ETH-4
PO-5	Deliver awareness resources and recovery-action tracking	All of FR55–FR60 implemented; checklists cover bank contact, password change, IDCARE, police and credit ban	FR55–FR60, ETH-2
PO-6	Deliver administration, analytics, de-identified export and immutable audit logging	All of FR67–FR72 implemented; exports pass the re-identification test in Section 11.4	FR67–FR72, ETH-4
PO-7	Research, propose and obtain approval for 15–20 enhanced functional requirements	Approved list signed off by supervisor and stakeholders before the Final SRS	Client brief; OI-4

4.3.2 Quality objectives

ID	Objective	Measure of success	Traceability
QO-1	Meet the performance targets	Pages load within 3 s; queries under 2 s at 100,000 records; load-tested toward 1,000 concurrent users	NFR-1–NFR-3
QO-2	Meet WCAG 2.1 Level AA on every page	Automated and manual accessibility audit passed with zero Level A or AA failures	NFR-13, C7
QO-3	Pass security verification	No high or critical findings open at handover; OWASP Top 10 checks documented	NFR-7–NFR-11
QO-4	Normalise the database to 3NF with enforced referential integrity	Schema review signed off; all relationships use foreign-key constraints	NFR-20, NFR-27
QO-5	Achieve full requirements traceability	Every FR traceable to design, code module and test case in the traceability matrix	Appendix D

4.3.3 Ethical and privacy objectives

ID	Objective	Measure of success	Traceability
EO-1	Collect only the personal data the service requires	Field-level justification register complete; every stored field mapped to a requirement	ETH-6, NFR-30
EO-2	Ensure no public output or export can re-identify a reporter	k-anonymity threshold enforced; re-identification test passed	ETH-4, FR50, FR71
EO-3	Ensure risk scoring is transparent and non-discriminatory	Scoring factors inspectable; no size, industry, demographic or locality attribute used as a scoring input	ETH-3
EO-4	Ensure results are never presented as professional certification	Advisory disclaimer present on every result and report screen	ETH-1, ETH-2
EO-5	Honour account and data deletion requests without destroying the audit trail	Deletion procedure implemented and tested per Section 11.4 (ER-7)	FR12, NFR-29, FR72

4.3.4 Project (delivery) objectives

ID	Objective	Measure of success
DO-1	Deliver all milestones M0–M16 on schedule	No milestone slips more than 5 working days without approved re-baseline

ID	Objective	Measure of success
DO-2	Deliver at zero cash cost during development	Total cash expenditure remains AUD \$0 through handover
DO-3	Post every iteration plan and report version to the unit forum	Complete forum record, Appendix C, with no missing version
DO-4	Maintain stakeholder engagement at the agreed cadence	Fortnightly supervisor checkpoint and monthly stakeholder review held and minuted
DO-5	Hand over a maintainable, documented system	Documentation complete; handover pack accepted by Council

4.4 Stakeholders

4.4.1 Stakeholder register

ID	Stakeholder	Category	Interest in the project	Influence	Power/Interest quadrant	Engagement strategy
S1	Hume City Council (as client body)	External, primary	Owns the service; accountable for community safety outcomes and for the personal data held	High	High power, high interest	Manage closely — monthly review, formal sign-off on every baseline
S2	CyberSafe Services manager	External, primary	Sponsors the project; owns operational delivery of the service	High	High power, high interest	Manage closely — primary client contact; approves scope changes
S3	Project supervisor / Lecturer	Academic, primary	Assesses the project; provides direction and approves enhanced FRs	High	High power, high interest	Manage closely — fortnightly checkpoint; approval gate at each milestone
S4	Council reviewers / analysts	External, primary user	Will operate the review queue daily; usability	Medium	Low power, high interest	Keep informed and consult — involved in usability testing

ID	Stakeholder	Category	Interest in the project	Influence	Power/Interest quadrant	Engagement strategy
			directly affects their workload			of the review workflow
S5	Council system administrators	External, user	Manage users, roles and categories; responsible for operational security	Medium	Medium power, high interest	Keep satisfied — consulted on RBAC, admin tooling and handover
S6	Residents of Hume	External, end user	Primary beneficiaries; check, report, learn and recover	Low individually	Low power, high interest	Keep informed — represented through Council; usability testing with community participants
S7	Small businesses, community organisations and NFPs	External, end user	Beneficiaries with distinct needs (invoice fraud, BEC, payment redirection)	Low	Low power, high interest	Keep informed — represented through Council; requirements validated with Council
S8	Scam victims (as a distinct group)	External, end user	Arrive distressed; need recovery guidance and non-judgemental treatment	Low	Low power, high interest	Design-led consideration — tone, language and ethical requirements written for them
S9	Group CyberKent (development team)	Internal, primary	Delivers the system; assessed on it	High	High power, high interest	Continuous — daily coordination, weekly internal review
S10	Council ICT and information governance	External	Must be satisfied on security, privacy, data	Medium	Medium power, medium interest	Keep satisfied — schema and security review

ID	Stakeholder	Category	Interest in the project	Influence	Power/Interest quadrant	Engagement strategy
			location and handover			before implementation
S11	Council privacy officer	External	Accountable for Privacy Act and Victorian PDP Act compliance	Medium	Medium power, high interest	Consult — sign-off required on the database ethics review (Section 11)
S12	Council communications team	External	Publishes alerts; owns Council's public voice and reputation	Low	Low power, medium interest	Keep informed — consulted on alert tone and publication workflow
S13	External referral bodies — Scamwatch/NASC, IDCARE, Victoria Police, ACSC	External, secondary	Receive referrals from the system; their guidance is surfaced in it	Low	Low power, low interest	Monitor — no integration this phase; referral links maintained accurately
S14	University / unit coordinator	Academic	Governs assessment and submission process	Medium	Medium power, medium interest	Keep informed — forum submissions per Appendix C
S15	Future maintainers	External, future	Inherit the codebase	None now	Low power, low interest	Serve through documentation — architecture, schema and API docs

4.4.2 Stakeholder analysis matrix

	Low interest	High interest
High power	(none)	S1 Council, S2 CyberSafe manager, S3 Supervisor, S9 CyberKent
Medium power	S14 Unit coordinator	S5 Administrators, S10 ICT/governance, S11 Privacy officer
Low power	S13 Referral bodies, S15 Future maintainers	S4 Reviewers, S6 Residents, S7 Organisations, S8 Victims, S12 Communications

The pattern that matters here: **the people who bear the consequences of the design have the least power over it**. Residents, victims and reviewers sit in the low-power, high-interest quadrant. That asymmetry is precisely why the ethical requirements (Section 11 and ETH-1 to ETH-6) are treated as binding constraints rather than aspirations — they are the mechanism by which the interests of stakeholders who cannot advocate for themselves are protected in the design.

4.5 Required project team

4.5.1 Team composition and roles

Group **CyberKent** operates with six roles. Because the group is small, each member holds one primary role and one secondary role, so that no capability is single-threaded.

Role	Primary responsibilities	Skills required	Member	Allocation
Project Manager	Schedule, milestones, risk register, communication plan, stakeholder liaison, forum submissions, scope control	Planning, risk management, stakeholder communication, MoSCoW prioritisation	<i>[Insert name and student ID]</i>	100%
Business Analyst / Requirements Lead	Requirements elicitation and baseline, traceability matrix, SRS authorship, open-issue register, enhanced FR research	Requirements engineering, SRS documentation, analysis, stakeholder interviewing	<i>[Insert name and student ID]</i>	100%
Solution Architect / Backend Lead	Architecture, API design, Express services, authentication and RBAC, security controls, integration	Node.js, TypeScript, Express, REST design, JWT, OWASP practices	<i>[Insert name and student ID]</i>	100%
Database Designer / Data Steward	Prisma schema, 3NF normalisation, migrations, indexing, retention and anonymisation design, owner of the database ethics review (Section 11)	PostgreSQL, Prisma, normalisation, data modelling, privacy-by-design	<i>[Insert name and student ID]</i>	100%

Role	Primary responsibilities	Skills required	Member	Allocation
Frontend / UI-UX Lead	React components, design system implementation, responsive layout, WCAG 2.1 AA compliance, usability testing	React, TypeScript, Tailwind, accessibility, interaction design	<i>[Insert name and student ID]</i>	100%
QA / Test Lead	Test strategy, unit/integration/API tests, accessibility and security verification, defect management, UAT coordination	Test design, Jest/Vitest, API testing, accessibility auditing	<i>[Insert name and student ID]</i>	100%

To be completed before submission: insert each member's full name and student ID. Roles, responsibilities and allocations above are agreed and final.

4.5.2 Secondary role coverage

Capability	Primary	Backup
Requirements and documentation	Business Analyst	Project Manager
Backend and API	Solution Architect	Database Designer
Database and migrations	Database Designer	Solution Architect
Frontend and accessibility	Frontend Lead	QA Lead
Testing and quality	QA Lead	Frontend Lead
Client and supervisor liaison	Project Manager	Business Analyst

4.5.3 Skills gap analysis and mitigation

Honest assessment of where the team is not yet strong, and what is being done about it.

Capability	Current level	Gap	Mitigation
React, TypeScript, Tailwind	Strong	None	Demonstrated — landing page and prototype delivered
Node.js, Express, REST	Good	Minor	Architecture already specified; patterns fixed in the development rules

Capability	Current level	Gap	Mitigation
PostgreSQL and normalisation	Good	Minor	Schema peer-reviewed before first migration
Prisma ORM	Moderate	Moderate	Two-week ramp-up scheduled ahead of M8; migration workflow proven on a spike
Security engineering (OWASP)	Moderate	Significant	Controls fixed by rules rather than by judgement; security checklist per pull request; external review requested at M14
Privacy law and de-identification	Low	Significant	Primary mitigation for this project. Section 11 written against APP and VPDSS obligations; Council privacy officer sign-off sought at M8
WCAG 2.1 AA auditing	Moderate	Moderate	Accessible component library (shadcn/ui); accessibility in the per-page Definition of Done; audit at M14
AI / ML (FastAPI, OCR, NLP)	Low	Deferred	Explicitly out of scope this phase; architecture reserves the integration point only

The two significant gaps — security engineering and privacy law — are the reason those areas are governed by **prescribed rules and external review** rather than by team judgement. This is a deliberate risk-transfer decision recorded as risk RK-08 and RK-09 in Section 9.

5.6 Constraints

ID	Constraint	Type	Source
C1	Core technologies must be free and open-source or free-tier. There is no licensing budget .	Financial	Client
C2	The architecture must be a modular monolith with service-oriented boundaries capable of staged migration to microservices	Technical	Architecture
C3	AI capability must live in a separate Python/FastAPI	Technical	Architecture

ID	Constraint	Type	Source
	service; the core system must be fully functional when AI is unavailable		
C4	Strict layer separation: controller → service → repository. No business logic in controllers or in the database	Technical	Standards
C5	Schema changes only through version-controlled Prisma migrations; database normalised to 3NF with foreign keys enforced	Technical	NFR-20, NFR-27
C6	Fixed security controls: bcrypt, JWT with refresh tokens, RBAC, Helmet, rate limiting, server-side validation on every request	Security	NFR-7 to NFR-11
C7	Every page must meet WCAG 2.1 Level AA and use mobile-first responsive design	Accessibility	NFR-13, NFR-14
C8	The UI must follow the approved design system without deviation	Design	D4
C9	Git feature-branch workflow, conventional commits, no direct commits to main, no secrets in source control	Process	Standards
C10	Data minimisation and privacy by design; personal data collected only where the service requires it	Ethical/legal	ETH-6, NFR-30
C11	Delivery within the academic calendar: 30	Schedule	University

ID	Constraint	Type	Source
	June to 27 November 2026		
C12	Team capacity is fixed at six part-time student members; no additional resource can be added	Resource	University
C13	Personal data must be handled in accordance with the Privacy Act 1988 (Cth), the Australian Privacy Principles and the Victorian Privacy and Data Protection Act 2014	Legal	Legislation

5.7 Assumptions

ID	Assumption	If it proves false
A1	Free-tier hosting (Vercel, Render, Neon, Cloudinary) provides sufficient capacity for initial deployment and demonstration	Paid tiers required before public promotion — costed in Section 7.2; recorded as risk RK-01
A2	An SMTP provider reachable by Nodemailer is available for verification and notification email	Notification requirements FR61–FR66 cannot be met; fallback to in-app only
A3	OpenStreetMap tile service remains available under its usage policy	Scam map degrades to a non-map regional summary
A4	Council can staff the reviewer and administrator roles that FR37–FR42 and FR49–FR51 require	The verification and alerting workflow cannot operate; recorded as risk RK-05
A5	Stakeholders and supervisor will approve the enhanced FRs before the Final SRS	Final SRS issues with the baseline only and the omission recorded
A6	Council will make a privacy officer available to review the database design at M8	Team proceeds with documented self-assessment and flags the unmitigated risk; recorded as risk RK-09
A7	External intelligence APIs and LLM providers are future dependencies only; no baseline requirement depends on them	No effect on this phase

ID	Assumption	If it proves false
A8	Requirements will remain stable following stakeholder confirmation of the open issues	Change control per Section 5.9 applies
A9	All six team members remain available for the project duration	Secondary role coverage (Section 4.5.2) activates; recorded as risk RK-06

6. Project milestones and schedule

6.1 Milestone schedule

Milestones are gates, not activities: each represents a verifiable deliverable state that must be reached before the next phase begins.

ID	Milestone	Deliverable evidence	Planned date	Dependency	Status
M0	Project initiation and client brief analysis complete	Brief analysed; team formed; roles assigned	30 Jun 2026	—	Complete
M1	Requirements baseline established	FR1–FR72, NFR-1–NFR-30, ETH-1–ETH-6 catalogued; ambiguities registered	8 Jul 2026	M0	Complete
M2	Technology stack selected and architecture baselined	D2, D3, D5 issued and peer-reviewed	12 Jul 2026	M1	Complete
M3	UI/UX design system baselined	D4 issued; component inventory defined	15 Jul 2026	M2	Complete
M4	Interim SRS Report v1.0 issued	D1 issued for review	16 Jul 2026	M1–M3	Complete
M5	Frontend prototype demonstrating the stack end to end	Landing page and interactive scam-check prototype running	31 Jul 2026	M3	Complete
M6	Interim Project Report issued and posted to forum	D6 (this document)	1 Aug 2026	M4, M5	Complete
M7	Enhanced functional requirements approved	D7 — 15–20 FRs signed off by supervisor and stakeholders	12 Aug 2026	M4	In progress
M8	Database design and ethics review signed off	D8 — Prisma schema, 3NF evidence, ethics	21 Aug 2026	M6, M7	Planned

ID	Milestone	Deliverable evidence	Planned date	Dependency	Status
		register, privacy officer sign-off			
M9	Modules 1–2 complete (authentication and access control)	FR1–FR12 implemented and tested	4 Sep 2026	M8	Planned
M10	Modules 3–4 complete (scam content, URL and contact analysis)	FR13–FR24 implemented and tested	18 Sep 2026	M9	Planned
M11	Modules 5–6 complete (reporting and evidence)	FR25–FR36 implemented and tested	2 Oct 2026	M9	Planned
M12	Modules 7–8 complete (review, verification, duplicate detection)	FR37–FR48 implemented and tested	16 Oct 2026	M11	Planned
M13	Modules 9–12 complete (alerts, awareness, notifications, admin and audit)	FR49–FR72 implemented and tested	30 Oct 2026	M11, M12	Planned
M14	System verification complete	D10 — accessibility audit, security verification, load test, UAT signed off	13 Nov 2026	M13	Planned
M15	Final SRS Report issued	D11 — baseline plus approved enhanced FRs, all open issues resolved	20 Nov 2026	M7, M14	Planned
M16	Deployment, handover and final presentation	D12 accepted by Council; final presentation delivered	27 Nov 2026	M14, M15	Planned

6.4 Schedule contingency

Total elapsed duration is 22 weeks. Approximately **2.5 weeks of float** is distributed across the implementation iterations (It-5 to It-9) rather than being held as a single block at the end, so that a slip in one module is absorbed locally rather than accumulating.

The MoSCoW priority ordering is the schedule's release valve. Modules M1, M2, M3, M4, M5, M6, M7 and M12 are rated **Must**; modules M8, M9, M10 and M11 are rated **Should**. If the schedule comes under pressure, the Should modules are reduced in depth before any Must module is compromised, and the reduction is agreed with stakeholders rather than absorbed silently.

8. Feasibility study

Feasibility is assessed across six dimensions. Each carries an assessment, the evidence for it, the risks identified and the conditions attached.

8.1 Technical feasibility

Assessment: FEASIBLE — HIGH CONFIDENCE

8.1.1 Technology availability and maturity

Every technology in the stack is mature, actively maintained, widely adopted and free. There is no component whose availability is in doubt and no component that requires a licence. React, TypeScript, Node.js, Express, PostgreSQL, Prisma, Tailwind and Docker are all industry-standard with large ecosystems and strong community support — which also means the team can resolve problems from public documentation rather than depending on vendor support the project cannot buy.

8.1.2 Complexity assessment by module

Module	Technical complexity	Assessment
M1 Authentication (FR1–FR6)	Low–Medium	Well-established patterns; JWT and bcrypt are standard. Main care needed on token handling and non-disclosure in error messages
M2 Profile and access control (FR7–FR12)	Low–Medium	RBAC middleware is a solved problem. Account deletion (FR12) is the complex part — see ER-7, Section 11
M3 Scam content analysis (FR13–FR18)	Medium–High	Rule-based keyword and pattern detection with a transparent scoring model. The difficulty is not the code — it is designing a scoring method that is explainable and non-discriminatory (ETH-3)
M4 URL and contact analysis (FR19–FR24)	Medium	Pattern detection against known suspicious URL characteristics; indicator matching against stored reports. Deliberately excludes external reputation APIs this phase
M5 Report management (FR25–FR30)	Low–Medium	Standard CRUD with state machine, validation and reference generation
M6 Evidence management (FR31–FR36)	Medium–High	Secure upload is where file-handling vulnerabilities live. Type/MIME validation, size limits, no execution, access control, access logging, EXIF stripping
M7 Review and verification (FR37–FR42)	Medium	Queue, assignment and state transitions with audit logging

Module	Technical complexity	Assessment
M8 Duplicate detection (FR43–FR48)	Medium–High	Text similarity comparison at scale is the performance-sensitive part; must meet NFR-3 at 100,000 records. Indexing strategy is critical
M9 Community alerts (FR49–FR54)	Medium	Workflow is straightforward; the anonymisation guarantee (FR50) is the hard requirement — see ER-2, Section 11
M10 Awareness and recovery (FR55–FR60)	Low	Content management, categorisation, search and checklist tracking
M11 Notifications (FR61–FR66)	Medium	Email delivery reliability and subscription management; requires retry handling
M12 Admin, analytics and audit (FR67–FR72)	Medium–High	The de-identified export (FR71) is the difficult requirement — see ER-2 and ER-3, Section 11. Immutable audit logging must resist tampering

Four modules carry Medium–High complexity. In three of them (M3, M9, M12) the difficulty is **ethical and analytical rather than technical** — writing the code is not the hard part; getting the scoring transparent, the anonymisation genuine and the export non-re-identifiable is. This is why the database ethics review (D8) is scheduled ahead of implementation.

8.1.4 Technical risks and mitigations

Risk	Mitigation
Team's Prisma experience is moderate	Two-week ramp-up before M8; migration workflow proven on a throwaway spike first
Duplicate detection may not meet NFR-3 at scale	Indexing strategy designed at M8; comparison scoped to candidate sets rather than full scans; performance-tested against a 100,000-record dataset
Secure file upload is a common vulnerability class	Controls fixed by rule (allowlist, MIME and extension check, size limit, no execution, EXIF stripping); code review focused on this module
Free-tier resource limits during development	Local Docker environment for development; free tiers used for demonstration only

Conclusion. The stack is proven, the architecture is designed, the frontend is demonstrated working, and the four hard modules have been identified with their difficulties named. Technically feasible.

8.2 Economic feasibility

Assessment: FEASIBLE — STRONG

The full analysis is in Section 7. Summarised:

Measure	Value
Cash cost to develop	AUD \$0
Notional development labour value	AUD \$64,800
Annual operating cost	≈ AUD \$28,400 (infrastructure \$4,400 + 0.2 FTE maintenance \$24,000)
Year 1 total including one-off costs	AUD \$43,400
Three-year TCO	AUD \$100,200
Break-even benefit required	≈ AUD \$28,400/yr — about 0.14% of estimated local reported scam losses
Benefit–cost ratio	3.5:1 (conservative) to 13.9:1 (optimistic)
Funding routes identified	Six (Section 7.3); primary route F2, Council community safety budget

The economic case is unusually robust because the break-even threshold is so low. The project does not need to be highly effective to be worth doing — it needs to prevent a small handful of significant losses per year. Section 7.4.2 shows the conclusion survives a five-fold error in the benefit estimate.

The one economic condition: free-tier hosting must be replaced with paid tiers before the service is publicly promoted at scale. This is a budget decision of approximately \$4,400 per year, and it must be approved before promotion rather than discovered afterward. Recorded as risk RK-01 and condition 1 in Section 8.7.

8.3 Operational feasibility

Assessment: FEASIBLE — CONDITIONAL

This is the dimension where the project is most dependent on something outside the team's control.

8.3.1 PIECES analysis

Dimension	Current state without the system	State with the system
Performance	No capability exists; residents wait for national channels or act on guesswork	Immediate self-service checking; reports triaged in a managed queue
Information	Council has no visibility of local scam activity	Local intelligence, regional trends, de-identified statistics
Economics	Losses are borne entirely by residents and organisations	Losses reduced; Council effort targeted rather than generic
Control	No control over scam exposure; no audit of anything	RBAC, immutable audit trail, moderated publication, access logging
Efficiency	Ad hoc, duplicated, reactive Council effort	Structured workflow, duplicate detection, reusable resources

Dimension	Current state without the system	State with the system
Services	No local scam service exists	Check, report, alert, educate and recover in one accessible portal

8.3.2 The operational dependency

The verification and alerting workflow (FR37–FR42, FR49–FR51) **requires trained Council staff**. A report that no one reviews is not a service; an alert that no one approves is never published. The estimate is approximately **0.4 FTE** of reviewer capacity, which Council would most plausibly meet by redeploying existing community safety staff rather than by new hiring.

This is assumption A4 and risk RK-05. It is the single condition on which operational feasibility rests, and it must be confirmed by Council before M13, when the review workflow is delivered. If Council cannot commit the capacity, the service can still deliver the self-service checking, awareness and recovery capabilities — but the reporting, verification and alerting capabilities, which are the core of the intelligence value, would not function.

8.3.3 Other operational factors

Factor	Assessment
User acceptance — community	Good. No installation, no cost, no login required for the public alerts, map and awareness content. Registration is required only to report
User acceptance — staff	Good, with training. The review queue must be efficient enough to be usable daily. Reviewers are involved in usability testing before M14
Training requirement	Low for residents (NFR-12 requires minimal training). Moderate for staff — administrator and reviewer guides plus a training session are in D12
Accessibility of the service	Designed in. WCAG 2.1 AA on every page; mobile-first from 360 px; plain language; no technical jargon in user-facing text
Change management	Low disruption. The system replaces nothing; there is no migration, no legacy process to retire and no existing user base to transition
Support model	0.2 FTE developer for application maintenance; Council ICT for operational support. Defined in the handover pack

8.4 Schedule feasibility

Assessment: FEASIBLE — TIGHT

Factor	Assessment
Total duration	22 weeks (30 Jun – 27 Nov 2026)
Elapsed at interim	4.5 weeks (20%)
Milestones complete at interim	7 of 17 (M0–M6)

Factor	Assessment
Progress against plan	On schedule — all interim milestones met on their planned dates
Float	~2.5 weeks distributed across implementation iterations
Critical path length	13 milestones
Parallelism available	One planned stream split (M10 and M11)

Why it is tight rather than comfortable. Twelve functional modules must be implemented, tested and deployed in the ten weeks between M8 and M14. That is achievable only because the design work is genuinely finished — architecture, technology, UI system and standards are all baselined, so implementation is not competing with design decisions. The team is not starting implementation with open architectural questions, which is the usual reason student projects overrun.

What protects the schedule. Three mechanisms:

23. **MoSCoW prioritisation.** Should-rated modules (M8–M11) are reduced in depth before any Must-rated module is compromised.
24. **Distributed float.** Slip is absorbed locally in the iteration where it occurs rather than accumulating to the end.
25. **Fixed standards.** Because the development rules prescribe the patterns, implementation time is not consumed by re-deciding how to structure each module.

What threatens it. M7 (enhanced FR approval) and M8 (database design sign-off) both depend on external parties — the supervisor, stakeholders and the Council privacy officer. Delay there is the most likely source of schedule loss, which is why both are scheduled early with the largest float allowance and are actively chased rather than waited on. Recorded as risks RK-02 and RK-09.

8.5 Legal and regulatory feasibility

Assessment: FEASIBLE — CONDITIONAL ON THE CONTROLS IN SECTION 11

The system holds personal information about people who have been targeted or defrauded, including financial loss detail and evidence files. That places it squarely within Australian privacy law, and the legal analysis below is the reason the database ethics review is a formal deliverable.

Instrument	Relevance	Compliance approach
Privacy Act 1988 (Cth) and the Australian Privacy Principles	Governs collection, use, disclosure, quality, security, access, correction and cross-border disclosure of personal information	APP 1 (open and transparent management) — privacy policy and collection notice; APP 3 (collection) — data minimisation per ER-1; APP 5 (notification) — collection notice at every collection point; APP 6 (use and disclosure) — purpose limitation enforced in design; APP 8 (cross-border) — data residency pinned to Australia per ER-15; APP 11 (security) — controls per NFR-7 to NFR-11; APP 12/13 (access

Instrument	Relevance	Compliance approach
		and correction) — FR7, FR8 and the dispute process in ER-13
Notifiable Data Breaches scheme	A breach involving this data would very likely meet the "serious harm" threshold, triggering mandatory notification to the OAIC and affected individuals	Incident response procedure to be documented in D12; audit logging (FR72) provides the forensic record; encryption and access control reduce the likelihood of a notifiable breach
Privacy and Data Protection Act 2014 (Vic) and the Information Privacy Principles	Applies to Victorian public sector organisations, including local councils	The APP controls above satisfy the IPPs; Council privacy officer sign-off at M8 confirms alignment
Victorian Protective Data Security Standards (VPDSS)	Sets the security standards expected of Victorian public sector information	Layered security architecture; RBAC; audit logging; encryption in transit and at rest; access review reporting
Disability Discrimination Act 1992 (Cth)	A public-facing government service must be accessible	WCAG 2.1 Level AA on every page (NFR-13); independent audit at M14
Defamation law	Publishing that a named person or business is a scammer carries real defamation exposure — contact indicators are frequently spoofed and often belong to innocent victims	Structural mitigation: the system never publishes the identity of an alleged scammer (X12); indicators are never publicly listed; alerts describe scam <i>patterns</i> , not <i>people</i> ; a dispute and correction process exists (ER-13)
Copyright	Awareness content may reproduce third-party material	Original content or properly licensed and attributed material only
Records management obligations	Council records have retention obligations	Retention schedule defined per table (ER-6), aligned with Council records policy at M8

The three legal conditions. Legal feasibility is conditional on: (1) the data-handling controls in Section 11 being implemented as designed rather than deferred; (2) Council privacy officer sign-off on the database design at M8; and (3) data residency being confirmed as Australian across all four hosting providers. None of these is difficult. All three are avoidable failures if left to the end.

8.6 Ethical and social feasibility

Assessment: FEASIBLE — WITH THE MITIGATIONS IN SECTION 11 TREATED AS BINDING

Ethical consideration	Assessment
Risk of causing fear	Manageable. ETH-1 prohibits fear-based presentation; tone, colour and result presentation are governed by the design system. A calm interface is a design requirement, not a preference
Risk of false reassurance	Manageable. ETH-2 requires an explicit statement on every result that the assessment is general guidance based on supplied information, cannot guarantee protection, and that serious incidents need qualified help
Risk of discriminatory scoring	Manageable but requires vigilance. ETH-3 prohibits discrimination by organisation size, industry or resources. Section 11 (ER-8, ER-9) extends this to prohibit demographic, locality and language attributes as scoring inputs, and requires review of the keyword list for proxies of nationality, ethnicity, religion or language
Risk of re-identifying reporters	The most serious ethical risk in the project. Addressed by ER-2, ER-3 and ER-4 in Section 11 with k-anonymity enforcement, quasi-identifier generalisation and separation of identity from incident data
Risk of harm to innocent third parties	Real and structurally mitigated. Phone numbers and email addresses in scam reports frequently belong to innocent people whose identity was spoofed. The system never publicly identifies them, stores indicators with a verification status and confidence, and provides a dispute process (ER-13)
Risk to vulnerable users	Addressed. Plain language, WCAG 2.1 AA, no judgemental language, recovery guidance rather than blame, and no collection of data about children beyond what the service requires (ER-14)
Transparency of automated decisions	Addressed. FR18 requires every score to display its contributing indicators in plain language. ETH-5 requires future AI output to be advisory, explainable and human-reviewed
Social equity	Positive. A free, accessible service reaches residents who cannot purchase commercial protection — this is a redistributive benefit, not a neutral one

9. Major risks

9.1 Risk assessment method

Risks are scored on a 5 × 5 matrix. **Likelihood** and **Impact** are each rated 1 (very low) to 5 (very high); **Risk score** is their product.

Score	Rating	Response required
1–4	Low	Accept and monitor
5–9	Moderate	Mitigate; review monthly

Score	Rating	Response required
10–14	High	Mitigate actively; review fortnightly; escalate to supervisor
15–25	Critical	Immediate action; escalate to stakeholders; may trigger re-baseline

Each risk carries a **trigger** — the observable signal that the risk is materialising — because a mitigation that is only activated after the damage is visible is not a mitigation.

9.2 Risk register

ID	Risk	Category	L	I	Score	Rating	Owner	Response	Mitigation	Trigger
RK-01	Free-tier hosting cannot meet NFR-2 (1,000 concurrent users) at public launch	Technical/Financial	4	3	12	High	Solution Architect	Mitigate	Load-test against NFR-2 before M16; architecture supports vertical then horizontal scaling; paid-tier cost quantified at ~\$4,400/yr (Section 7.2.2) and raised as feasibility condition 1	Load test fails, or response times degraded above 2 concurrent sessions
RK-02	Enhanced FR approval (M7) is delayed, blocking the Final SRS and compressing implementation	Schedule/Stakeholder	3	4	12	High	Business Analyst	Mitigate	Candidate list prepared in advance from architecture extension points; submitted early with justification and cost per item; chased at each fortnightly checkpoint	No decision received by 12 Aug 2024
RK-03	Scope growth from enhanced FRs squeezes implementation time	Scope	3	4	12	High	Project Manager	Mitigate	Cap at the approved 15–20; each candidate assessed for schedule feasibility before proposal; MoSCoW prioritisation applied; Must modules built first	Approvals exceed 10 items, or item requires a new external dependency

ID	Risk	Category	L	I	Score	Rating	Owner	Response	Mitigation	Trigger
RK-04	Evidence handling exposes personal data — files contain more personal information than reporters realise (EXIF location, bank details in screenshots, third-party identities)	Security/Privacy	3	5	15	Critical	Database Designer	Mitigate	RBAC access restriction (FR35); access logging (FR36); EXIF stripping on ingest (ER-5); allowlist type and MIME validation; size limits; no execution; Cloudinary secure URLs; database stores metadata and URLs only; encryption at rest	Any unauthorised access recorded; the evidence access any file stored without stripping
RK-05	Council cannot commit reviewer capacity, so reports are never verified and alerts are never published	Operational	3	4	12	High	Project Manager	Mitigate/Transfer	Capacity requirement (~0.4 FTE) quantified early and raised as feasibility condition 2; review workflow designed for efficiency; queue prioritisation reduces per-report time	No commitment confirmed M13
RK-06	Loss of a team member (withdrawal, illness, unavailability)	Resource	3	4	12	High	Project Manager	Mitigate	Every capability has a named backup (Section 4.5.2); no single-threaded knowledge; all decisions and designs documented in the repository, not held individually; code review means at least two people	Any member misses consequences stand-up without notice

ID	Risk	Category	L	I	Score	Rating	Owner	Response	Mitigation	Trigger
									have read every module	
RK-07	Duplicate detection (FR43–FR48) fails NFR-3 performance at 100,000 records	Technical/Performance	3	3	9	Moderate	Solution Architect	Mitigate	Indexing strategy designed at M8; similarity comparison scoped to candidate sets selected by indexed indicator match rather than full-table scan; performance-tested against a synthetic 100,000-record dataset before M13	Query time exceeds in performance testing
RK-08	Security vulnerability introduced through inexperience — the team's security engineering capability is a known gap	Security	3	5	15	Critical	Solution Architect	Mitigate/Transfer	Security controls prescribed by rule rather than left to judgement (bcrypt, JWT, RBAC, Helmet, rate limiting, Zod validation, parameterised queries via Prisma); security checklist on every pull request; no self-merge; external penetration test requested at M14	Any security finding in code review; any cheater bypasses meet a deadline
RK-09	Privacy officer review (M8) unavailable or delayed, so the database is built without governance sign-off	Legal/Schedule	3	5	15	Critical	Database Designer	Mitigate	Review requested at M6 with 3 weeks' notice; the ethics register (Section 11) is prepared in full so the review is a check rather than an	No confirmation review completed by 14 April 2026

ID	Risk	Category	L	I	Score	Rating	Owner	Response	Mitigation	Trigger
									analysis; controls designed in regardless of review timing	
RK-10	Re-identification of a reporter from a published alert or de-identified export	Ethical/Privacy	2	5	10	High	Database Designer	Mitigate	Anonymisation enforced at the data layer, not the presentation layer (ER-2); k-anonymity threshold with small-cell suppression on all exports and public views (ER-3); quasi-identifier generalisation — date to week, location to region; separation of identity from incident data (ER-4); re-identification test in the acceptance criteria for D8	Any exp public v returnin cell belo the k thresho
RK-11	Discriminatory or biased risk scoring — the scoring model or keyword list encodes proxies for nationality, ethnicity, language or socio-economic status	Ethical	3	4	12	High	Business Analyst	Mitigate	Prohibited-input list enforced in the schema (ER-8); keyword list reviewed for proxies before each release (ER-9); every score stored with its contributing factors and the keyword-set version, making bias auditable after the fact; ETH-3 compliance tested	Review identifie keyword function as a pro for a protecte attribute compla unfair scoring
RK-12	A third party is wrongly implicated — a	Ethical/Legal	4	4	16	Critical	Business Analyst	Mitigate	Structural: the system never publicly	Any compla from a

ID	Risk	Category	L	I	Score	Rating	Owner	Response	Mitigation	Trigger
	phone number or email address in a report belongs to an innocent person whose identity was spoofed								identifies alleged scammers (X12); indicators are never publicly listed; alerts describe patterns not people; every indicator stored with verification status and confidence (ER-13); dispute and correction process defined	person whose contact detail appears the system any alert draft name an individual
RK-13	Accessibility work is deferred to the end and WCAG 2.1 AA is not achieved	Quality/Legal	3	3	9	Moderate	Frontend Lead	Mitigate	Accessibility is in the per-page Definition of Done, not a final phase; accessible component library (shadcn/ui) used throughout; automated checks in the build; independent audit at M14	Any page merged without keyboard navigation and confirmation
RK-14	External service outage — email provider, map tiles or file storage becomes unavailable	External/Operational	3	2	6	Moderate	Solution Architect	Mitigate	Graceful degradation designed in: AI unavailable → core reporting works; email down → retry from a background queue; map tiles unavailable → non-map regional summary; each integration wrapped in a service class with timeouts,	Monitor alert, or rate on external above threshold

ID	Risk	Category	L	I	Score	Rating	Owner	Response	Mitigation	Trigger
									retries and fallbacks	
RK-15	Requirements instability — stakeholders change direction after the baseline is set	Scope/Stakeholder	2	4	8	Moderate	Business Analyst	Mitigate	Formal baseline issued and reviewed; ambiguities raised as open issues rather than assumed; change control per Section 5.9; fortnightly checkpoints surface drift early	Any change request affecting Must-ra module M9
RK-16	Data residency is outside Australia on one or more hosting providers, breaching APP 8 expectations	Legal	2	4	8	Moderate	Solution Architect	Mitigate	Region pinned to Australia at provisioning for all four providers; data location register maintained; verified at M8 as feasibility condition 4	Provision confirm non-Australia region for any provision holding personal data
RK-17	Scope of the client brief remains ambiguous (OI-1) and the team builds the wrong product	Scope/Stakeholder	2	5	10	High	Business Analyst	Mitigate	Ambiguity identified during analysis and formally raised rather than silently resolved; the team's reading (scam detection and reporting) is supported by the title, all 14 modules and all 72 FRs; confirmation actively sought	No confirm received M8

9.5 Risk management process

Activity	Frequency	Owner	Output
Risk review	Fortnightly, at iteration close	Project Manager	Updated register; changed scores recorded

Activity	Frequency	Owner	Output
Critical risk review	Weekly	Risk owner	Status against mitigation; trigger check
New risk identification	Continuous — any team member may raise a risk at any time	All	Entry added with initial assessment
Escalation to supervisor	Any risk reaching High or Critical, or any trigger fired	Project Manager	Raised at the fortnightly checkpoint or immediately if Critical
Escalation to stakeholders	Any Critical risk with a fired trigger, or any risk requiring a scope, schedule or budget decision	Project Manager	Written notification within 2 working days

10. Communication management plan

10.1 Purpose

Communication on this project has to do three things: keep stakeholders informed at a level of detail appropriate to their role; get decisions made when the project is blocked on them; and leave a record. The third matters more than it might seem — several of the project's risks (RK-02, RK-05, RK-09, RK-17) materialise as *silence* from an external party, and a communication plan that does not create a record of what was asked and when cannot distinguish a decision that was declined from one that was never seen.

10.2 Communication matrix

#	Audience	Information communicated	Purpose	Frequency	Channel	Format	Owner
CM-1	Project supervisor / Lecturer	Progress against milestones, blockers, decisions needed, deliverable drafts	Direction, approval, assessment	Fortnightly (iteration close)	Scheduled meeting + written summary	Iteration report + agenda	Project Manager
CM-2	Project supervisor / Lecturer	Iteration plans and all report versions	Assessment evidence; visibility of individual	Every iteration and every version	Unit discussion forum	Posted link to the document	Project Manager

#	Audience	Information communicated	Purpose	Frequency	Channel	Format	Owner
			and group work				
CM-3	CyberSafe Services manager (client)	Milestone status, scope changes, decisions required, risk escalations	Client governance and approval	Monthly , plus on demand	Meeting (video or on-site) + written summary	Status report	Project Manager
CM-4	Hume City Council stakeholders	Formal deliverables for review and sign-off	Baseline approval	At each milestone gate	Email with attached deliverable	Formal document + sign-off sheet	Project Manager
CM-5	Council privacy officer	Database design, ethical risk register, data classification, retention schedule	Privacy governance sign-off (feasibility condition 3)	At M8 , then on material change	Scheduled review meeting	D8 review pack	Database Designer
CM-6	Council ICT and information governance	Architecture, security controls, data residency, hosting topology, handover pack	Technical governance and handover acceptance	M8 and M16	Meeting + documentation	Architecture and handover packs	Solution Architect
CM-7	Council reviewers and administrators	Workflow walkthroughs, usability testing, training	Ensure the system is usable by the people who must operate it	M12 (usability), M14 (UAT), M16 (training)	Workshop / testing session	Session + guides	QA Lead
CM-8	Team internal — stand-up	Progress since last, plan until next, blockers	Coordination	3× weekly, 15 min	Video call; asynchronous written update if absent	Verbal + written log	Project Manager
CM-9	Team internal — iteration review	Increment demonstration, retrospective, next-iteration commitments	Planning and quality	Fortnightly, 60 min	Video call	Iteration report	Project Manager

#	Audience	Information communicated	Purpose	Frequency	Channel	Format	Owner
CM-10	Team internal — code review	Implementation quality, standards compliance, security checklist	Quality gate	Every pull request	GitHub pull request	Review comments	All developers
CM-11	Team internal — decision log	Decisions, options considered, rationale	Traceability; prevents re-litigating settled questions	On every decision	Repository document	Decision record entry	Project Manager
CM-12	All stakeholders	Critical risk with a fired trigger	Immediate awareness and decision	Within 2 working days of the trigger	Email, escalating to phone	Risk escalation notice	Project Manager

10.3 Meeting schedule

Meeting	Participants	Cadence	Duration	Standing agenda
Internal stand-up	CyberKent team	3× weekly	15 min	Done / next / blocked
Iteration review and retrospective	CyberKent team	Fortnightly	60 min	Demonstration, retrospective, next commitments, risk review
Supervisor checkpoint	Team + supervisor	Fortnightly	30–45 min	Milestone status, decisions required, deliverable review, risks at High or above
Client status meeting	PM + BA + CyberSafe manager	Monthly	45 min	Milestone status, scope, risks, decisions, conditions
Milestone gate review	Team + supervisor + client	At each milestone	60 min	Deliverable walkthrough, acceptance decision, next gate
Privacy and governance review	Database Designer + PM + privacy officer + ICT	M8, M16	90 min	Data model, ethical risk register, retention, residency, sign-off

10.4 Escalation path

Level	Trigger	Escalate to	Response time
1	Issue blocking one team member	Role backup, then Project Manager	Same day
2	Issue blocking an iteration commitment	Project Manager, resolved within the team	1 working day
3	Decision required outside the team's authority; risk reaching High	Project supervisor	2 working days
4	Scope, schedule, budget or resource decision; risk reaching Critical	CyberSafe Services manager (client)	3 working days
5	Privacy, legal or data-breach concern	Council privacy officer and supervisor, simultaneously	Immediate — same day

Level 5 bypasses the sequence deliberately. A privacy concern that waits two working days for a supervisor checkpoint is a privacy concern that has been left unaddressed for two working days.

10.5 Document control and version management

Rule	Detail
Version numbering	0.x for drafts, x.0 for issued baselines, x.y for revisions to an issued baseline
Revision history	Every document carries a revision history table recording version, date, description and author
Single source of truth	All project documents live in the project repository; no document is authoritative in email or chat
Change to a baseline	Requires change control per Section 5.9; results in a new version number and a re-issue
Forum posting	Every iteration plan and every report version is posted to the unit discussion forum on issue — see Appendix C
Naming convention	<Document-Name>-v<version>.<ext>

10.7 Communication tools

Purpose	Tool
Formal client and supervisor communication	Email (creates the record)
Meetings	Video conferencing, with written summary circulated afterward

Purpose	Tool
Team coordination	Group messaging for day-to-day; nothing decided in chat without a decision-log entry
Source and code review	GitHub — repository, pull requests, issues
Document repository	Project repository, version-controlled
Assessment submission	Unit discussion forum
Task and progress tracking	Shared project tracker aligned to the iteration plan

11. Ethical risks in database design and mitigation actions

11.3 Data classification

Every field in the schema is classified. The classification determines its storage, access and retention rules.

Class	Definition	Examples	Storage rule	Access rule
C1 — Public	Intended for publication	Published alerts, awareness resources, scam categories, regions	Standard columns	No authentication required
C2 — Internal	Operational, non-personal	Report status, severity, category assignment, review queue state	Standard columns	Authenticated staff
C3 — Confidential	Personal information	Name, email address, notification preferences, report descriptions	Encrypted at rest; RBAC-restricted	Owner and authorised staff only
C4 — Sensitive	Personal information whose disclosure would cause serious harm	Financial loss amounts, evidence files and metadata, victim narratives, audit log	Encrypted at rest; RBAC + purpose-bound access; every access logged	Owner and assigned staff only, on a need-to-know basis

Class	Definition	Examples	Storage rule	Access rule
		contents, contact indicators of third parties		

11.4 Ethical risk register — database design mitigations

ID	Ethical risk	Severity	Primary mitigation	Verified by
ER-1	Over-collection of personal data	High	Field-level collection justification register; prohibited attribute list	Register complete
ER-2	Re-identification from published alerts	Critical	Anonymisation as data transformation; no FK from alert to report; pre-publication scan	Re-identification test
ER-3	Re-identification from de-identified export	Critical	k-anonymity ($k \geq 5$); small-cell and complementary suppression; view-level enforcement	Automated cell-count test
ER-4	Linkage attacks across tables	High	UUID keys; identity/incident separation; pseudonymous reporter token; role-scoped joins	Schema review
ER-5	Hidden personal data in evidence files	Critical	EXIF stripping on ingest; metadata allowlist; C4 classification; upload guidance	Automated EXIF test
ER-6	Indefinite retention	High	Per-table retention schedule; retention_until; automated purge; de-identify over delete	Schedule complete; purge tested
ER-7	Erasure vs immutable audit conflict	Critical	Crypto-shredding with UUID tombstone; de-identify reports; delete evidence; plain-language notice	End-to-end deletion test
ER-8	Discriminatory scoring inputs	Critical	Prohibited-input list; separate factor table; score content not people; bias review	Code and schema review
ER-9	Bias in the scam keyword table	High	Keyword review test; prohibited term classes; versioning; two-person sign-off; rationale field	Keyword review documented
ER-10	Excessive staff privilege / insider browsing	High	Purpose-bound access; row-level control; read logging; monthly access review	Access-review report

ID	Ethical risk	Severity	Primary mitigation	Verified by
ER-11	Personal data in the audit log	High	Metadata-only logging; hashes not values; prohibited-content list; C4 classification	Log schema review
ER-12	Backups and non-production copies	High	No production data outside production; synthetic seed data; encrypted backups; 35-day window	Seed script; backup config
ER-13	Innocent third parties as indicators	Critical	No public indicator listing; verification status and confidence; dispute process; expiry; no naming	Dispute process tested
ER-14	Data about children and vulnerable people	High	Minimum age 16; no vulnerability attributes; no child-specific collection	Schema review
ER-15	Cross-border data location	High	Australian region pinned; data location register; verified before first production record	Location register

Six risks are rated Critical: ER-2, ER-3, ER-5, ER-7, ER-8 and ER-13. Each is a design decision that must be made correctly before the first migration, because each becomes materially harder — and in the cases of ER-2, ER-3 and ER-7, close to impossible — to correct once real personal data exists in the database.

11.6 Governance and assurance

Control	Detail
Owner	The Database Designer holds the Data Steward role and is accountable for this register
Design gate	No Prisma migration is written until this register is reflected in the schema design
Independent review	Council privacy officer and Council ICT governance review and sign off at M8 (feasibility condition 3)
Verification	Every mitigation above carries a stated verification method; all are acceptance criteria for deliverable D8
Change control	Any schema change adding a column holding personal data re-triggers the ER-1 justification and a review of ER-4 and ER-6
Ongoing assurance	Access-review report monthly (ER-10); keyword bias review each release (ER-9); export log reviewed quarterly (ER-3)
Incident response	Any suspected breach escalates immediately under communication level 5 (Section 10.4) and is assessed against the Notifiable Data Breaches threshold within 24 hours

13. Current project status and next steps

13.1 Status summary

Dimension	Status
Schedule	On track. 7 of 17 milestones complete (M0–M6), all on their planned dates
Elapsed duration	4.5 of 22 weeks (20%)
Scope	Baselined , with 6 open issues before stakeholders and 8 scope changes recorded
Budget	AUD \$0 spent , as planned. No cash expenditure forecast before M16
Risks	4 Critical, 8 High, 5 Moderate — all with mitigations assigned and owners named
Team	6 members, all roles filled, all capabilities covered with named backups
Quality	Design artefacts peer-reviewed and internally consistent; frontend prototype demonstrates the stack working

13.2 What is complete

- **Requirements baseline** — 72 FRs across 12 modules, 30 NFRs, 6 ethical requirements, all specified, uniquely identified and traceable to the client brief.
- **Technology stack** — selected, justified layer by layer, and verified as satisfying all 30 NFRs on entirely free or free-tier technology.
- **System architecture v1.0** — modular monolith with service-oriented boundaries, layered security pipeline, defined processing pipelines, AI Gateway boundary and a four-phase scalability roadmap.
- **UI/UX design system** — visual hierarchy, colour system, typography, motion rules, component inventory, responsive breakpoints and accessibility rules.
- **Engineering standards** — development rules, prohibited practices, security ruleset, the ranked priority order and the Definition of Done, all adopted by the team.
- **Interim SRS Report** — issued 16 July 2026.
- **Frontend prototype** — public landing page and interactive scam-check prototype built and running, with the reusable component library and accessibility hooks implemented. This is the proof that the design is buildable.
- **Interim Project Report** — this document.

13.3 What is in progress

- **Enhanced functional requirements (D7, M7).** Eight candidate areas identified from the architecture's reserved extension points. Each is being assessed for justification, implementation cost and schedule feasibility ahead of submission for approval by 12 August 2026.

13.4 Immediate next steps

Order	Action	Owner	By
1	Post this report and all prior versions to the unit forum (Appendix C)	Project Manager	1 Aug 2026
2	Confirm the Council privacy officer review date for M8	Database Designer	8 Aug 2026
3	Submit the enhanced FR proposal for approval	Business Analyst	12 Aug 2026
4	Chase resolution of open issues OI-1, OI-2 and OI-3	Business Analyst	12 Aug 2026
5	Complete the Prisma ramp-up spike	Database Designer	14 Aug 2026
6	Design the data model with every Section 11 mitigation built in	Database Designer	19 Aug 2026
7	Verify Australian data residency across all four hosting providers	Solution Architect	19 Aug 2026
8	Database design and ethics review signed off (M8)	Database Designer	21 Aug 2026
9	Begin implementation of Modules 1 and 2 (authentication and access control)	Solution Architect	25 Aug 2026

13.5 Decisions required from stakeholders

#	Decision required	From	Needed by
1	Confirm the scope interpretation (OI-1)	CyberSafe Services manager	M8
2	Sign off the technology deviation (OI-2)	CyberSafe Services manager	M8
3	Confirm the NFR-30 restatement (OI-3)	CyberSafe Services manager	M8
4	Approve the enhanced functional requirements (OI-4)	Supervisor and stakeholders	M7
5	Sign off the hardware and software recommendations (OI-5)	CyberSafe Services manager	M8

#	Decision required	From	Needed by
6	Approve the proposed quantitative parameters (OI-6)	CyberSafe Services manager	M15
7	Confirm privacy officer availability for the M8 review (condition 3)	Council	8 Aug 2026
8	Confirm ~0.4 FTE reviewer capacity (condition 2)	CyberSafe Services manager	M13
9	Approve ~AUD \$4,400/yr paid hosting budget before public promotion (condition 1)	CyberSafe Services manager	M16

14. Conclusion

Group **CyberKent** is building **CyberNova**, an Online Scam Detection and Reporting System for Hume City Council CyberSafe Services, addressing a problem that is real, local, measurable and currently unaddressed: residents and small organisations in Hume have no trusted local way to check whether something is a scam, no easy way to report one, and no source of timely warnings — while Council has no visibility of what is happening to its own community.

At the interim point, the analysis and design phases are complete, internally consistent and peer-reviewed. Seventy-two functional requirements, thirty non-functional requirements and six ethical requirements are baselined. The technology stack, architecture, design system and engineering standards are all fixed. A working frontend prototype demonstrates the stack end to end, which means implementation begins from a proven foundation rather than an untested plan. All seven interim milestones were met on their planned dates, and nothing has been spent.

The feasibility study returns a **PROCEED recommendation across all six dimensions**, subject to five stated conditions. The economic case is the strongest element: the service needs to prevent roughly 0.14% of estimated local scam losses to cover its full recurring cost, and that conclusion survives a five-fold error in the underlying estimate. The operational case is the most dependent on others — it rests on Council committing approximately 0.4 FTE of reviewer capacity. The schedule is achievable but tight, and it is protected by distributed float, MoSCoW prioritisation and the fact that design decisions are already settled.

The most significant finding of the interim phase is one the team did not anticipate at proposal stage. This project's ethical risk does not live in the interface. It lives in the database — in what is stored, how it is linked, who can reach it, how long it is kept and what is let out. A schema that links a published alert back to its reporter makes anonymisation a display convention rather than a fact. An audit log that

stores request payloads makes the right to erasure unachievable. A scoring engine that can reach a suburb column makes non-discrimination a matter of developer restraint. Fifteen ethical risks in the data model have been identified, six of them critical, and each has been assigned concrete, testable mitigations that must exist in the schema before the first migration is written. That work — deliverable D8 at milestone M8, with Council privacy officer sign-off — is the immediate next step and the one condition in the feasibility recommendation that cannot be allowed to slip.

The team is on schedule, on budget, clear about what it does not yet know, and explicit about what it needs from stakeholders to proceed. The requirements baseline, the design, the risk position and the ethical analysis are all now before the Lecturer, the supervisor and Council for review, and the team will respond point by point to the feedback that comes back.

Appendix A — Glossary

Term	Definition
3NF	Third Normal Form — the database normalisation level required by NFR-20
ACCC	Australian Competition and Consumer Protection Commission
ACSC	Australian Cyber Security Centre
APP	Australian Privacy Principles, under the Privacy Act 1988 (Cth)
Audit log	An immutable record of security-relevant actions: who, what, when
BEC	Business Email Compromise — a scam type targeting organisations
Crypto-shredding	Rendering data unrecoverable by overwriting or destroying the means to read it, rather than deleting the record
DXA / EXIF	EXIF: metadata embedded in image files, often including GPS location and device identifiers
Evidence	Files attached to a scam report to substantiate it — screenshots, documents
FR / NFR / ETH	Functional Requirement / Non-Functional Requirement / Ethical Requirement
FTE	Full-Time Equivalent — a unit of staffing capacity
Indicator	An identifying artefact of a scam: a URL, phone number, email address or textual pattern
JWT	JSON Web Token — a stateless authentication token format
k-anonymity	A de-identification property: every released record is indistinguishable from at least $k-1$ others

Term	Definition
MoSCoW	Prioritisation scheme: Must have, Should have, Could have, Won't have this time
NASC	National Anti-Scam Centre
NDB scheme	Notifiable Data Breaches scheme, under the Privacy Act 1988 (Cth)
NFP	Not-for-profit organisation
NLP / OCR	Natural Language Processing / Optical Character Recognition — deferred AI capabilities
ORM	Object-Relational Mapping — here, Prisma
PIECES	An operational analysis framework: Performance, Information, Economics, Control, Efficiency, Services
Pseudonymisation	Replacing identifying data with a token that cannot identify a person without separately held information
Quasi-identifier	An attribute that is not identifying alone but becomes identifying in combination — postcode, date, amount
RACI	Responsible, Accountable, Consulted, Informed
RBAC	Role-Based Access Control
Risk score / classification	The numeric score and the band derived from it (Low, Medium, High) expressing scam likelihood
RPO / RTO	Recovery Point Objective / Recovery Time Objective — set by NFR-6 at 24 hours and 4 hours
SRS	Software Requirements Specification
TCO	Total Cost of Ownership
UAT	User Acceptance Testing
VPDSS	Victorian Protective Data Security Standards
WCAG 2.1 AA	Web Content Accessibility Guidelines 2.1, conformance Level AA

Appendix B — RACI matrix

R = Responsible (does the work) · **A** = Accountable (owns the outcome) · **C** = Consulted · **I** = Informed

Activity / Deliverable	Project Manager	Business Analyst	Solution Architect	Database Designer	Frontend Lead	QA Lead	Supervisor	Client
Project planning and schedule	A/R	C	C	C	C	C	I	I
Requirements baseline (D1)	C	A/R	C	C	I	C	C	C
Technology selection (D2)	I	C	A/R	C	C	I	I	I
System architecture (D3)	I	C	A/R	C	C	C	I	I
UI/UX design system (D4)	I	C	C	I	A/R	C	I	I
Engineering standards (D5)	C	I	A/R	C	C	C	I	—
Interim Project Report (D6)	A/R	R	C	C	C	C	C	I
Enhanced FRs (D7)	C	A/R	C	C	C	C	C	C
Database design and ethics review (D8)	C	C	C	A/R	I	C	C	C
Backend implementation (D9)	I	C	A/R	R	I	C	I	I
Frontend implementation (D9)	I	C	C	I	A/R	C	I	I
Test strategy and evidence (D10)	I	C	C	C	C	A/R	I	C
Final SRS Report (D11)	C	A/R	C	C	C	C	C	C
Deployment and handover (D12)	A	I	R	R	C	C	I	C
Risk management	A/R	C	C	C	C	C	I	I

Activity / Deliverable	Project Manager	Business Analyst	Solution Architect	Database Designer	Frontend Lead	QA Lead	Supervisor	Client
Stakeholder communication	A/R	R	I	I	I	I	I	I
Forum submissions	A/R	C	I	I	I	I	I	—

Appendix C — Document register and forum submission record

C.1 Project document register

Document	Role	Version	Date	Status
Requirements.md	Client brief — modules, FR1–FR72, NFRs, ethics [R1]	—	30 Jun 2026	Baselined
TechStack.md	Technology selection and rationale [R2]	1.0	12 Jul 2026	Baselined
System-Architecture.md	Architecture proposal [R3]	1.0	12 Jul 2026	Baselined
UI-Design.md	Visual design system [R4]	1.0	15 Jul 2026	Baselined
Rules.md	Development standards [R5]	1.0	12 Jul 2026	Adopted
Avoid.md	Prohibited practices and anti-patterns [R6]	1.0	12 Jul 2026	Adopted
Interim-SRS-Report.md	Interim SRS Report (D1)	1.0	16 Jul 2026	Issued for review
Interim-Project-Report	Interim Project Report (D6) — this document	1.0	1 Aug 2026	Issued for review
Final-SRS-Report.md	Final SRS Report (D11)	Draft	—	In preparation

C.2 Forum submission record

The assessment requires links to all versions of iteration plans and reports to be posted to the unit discussion forum, evidencing both individual and group work. The complete record is maintained below and reconciled against the forum at each iteration close.

#	Document and version	Date issued	Date posted	Contributing authors	Forum link
1	Iteration 1 Plan v1.0	30 Jun 2026	30 Jun 2026	Project Manager, Business Analyst	<i>[Insert forum link]</i>
2	Iteration 2 Plan v1.0	14 Jul 2026	14 Jul 2026	Project Manager	<i>[Insert forum link]</i>
3	Technology Stack v1.0	12 Jul 2026	12 Jul 2026	Solution Architect	<i>[Insert forum link]</i>
4	System Architecture v1.0	12 Jul 2026	12 Jul 2026	Solution Architect	<i>[Insert forum link]</i>
5	UI/UX Design System v1.0	15 Jul 2026	15 Jul 2026	Frontend / UI-UX Lead	<i>[Insert forum link]</i>
6	Interim SRS Report v1.0	16 Jul 2026	16 Jul 2026	Business Analyst, Solution Architect, PM	<i>[Insert forum link]</i>
7	Iteration 3 Plan v1.0	28 Jul 2026	28 Jul 2026	Project Manager	<i>[Insert forum link]</i>
8	Interim Project Report v1.0	1 Aug 2026	1 Aug 2026	All members — see contribution table below	<i>[Insert forum link]</i>

To be completed before submission: insert the forum URL for each posted item.

C.3 Individual contribution to this report

Section	Primary author	Contributing reviewers
1 Executive summary	Project Manager	All
2 Project background	Business Analyst	Project Manager
3 Problem statement	Business Analyst	Project Manager, Frontend Lead
4 General description, stakeholders, team	Project Manager	Business Analyst
5 Updated scope statement	Business Analyst	Project Manager, Solution Architect
6 Milestones and schedule	Project Manager	All

Section	Primary author	Contributing reviewers
7 Finance and funding	Project Manager	Business Analyst
8 Feasibility study	Project Manager, Solution Architect	Database Designer, QA Lead
9 Risks	Project Manager	All
10 Communication plan	Project Manager	Business Analyst
11 Ethical risks in database design	Database Designer	Solution Architect, Business Analyst
12 Responses to feedback	Project Manager	Business Analyst
13 Status and next steps	Project Manager	All
Appendices	Business Analyst	Project Manager

Appendix D — Requirements summary

D.1 Functional requirements by module

Module	FR range	Count	Priority
M1 User registration and authentication	FR1–FR6	6	Must
M2 User profile and access control	FR7–FR12	6	Must
M3 Scam content analysis	FR13–FR18	6	Must
M4 URL and contact analysis	FR19–FR24	6	Must
M5 Scam report management	FR25–FR30	6	Must
M6 Evidence management	FR31–FR36	6	Must
M7 Report review and verification	FR37–FR42	6	Must
M8 Duplicate and related report detection	FR43–FR48	6	Should
M9 Community scam alerts	FR49–FR54	6	Should
M10 Scam awareness and recovery	FR55–FR60	6	Should
M11 Notification and subscription	FR61–FR66	6	Should
M12 Administration, reporting and audit	FR67–FR72	6	Must
Total	FR1–FR72	72	

D.2 Non-functional requirements by attribute

Attribute	NFR range	Count
Performance	NFR-1–3	3
Reliability	NFR-4–6	3
Security	NFR-7–11	5
Usability	NFR-12–17	6
Maintainability	NFR-18–20	3
Scalability	NFR-21–24	4
Availability	NFR-25–26	2
Data integrity	NFR-27–28	2
Compliance and privacy	NFR-29–30	2
Total		30

D.3 Ethical requirements

ID	Requirement
ETH-1	The system shall avoid creating fear and shall not present results as professional cybersecurity certification
ETH-2	The system shall state clearly that results rest on supplied information, are general guidance, cannot guarantee protection, and that serious incidents may require a qualified professional or the relevant authority
ETH-3	Risk scoring shall be transparent and explainable, and shall not discriminate on grounds of organisation size, industry or available resources
ETH-4	Reporter identity shall be anonymised in every public output and de-identified export
ETH-5	Future AI output shall be advisory, explainable and human-reviewed before informing an administrative decision
ETH-6	Data collection shall be limited to what the service requires, following privacy by design

D.4 Traceability summary — module to architecture to ethics

Module	Main architecture components	NFR dependencies	Ethical risks (Section 11)
M1 Authentication	AuthenticationService, JWT, bcrypt, Nodemailer	NFR-7–10	ER-1, ER-14

Module	Main architecture components	NFR dependencies	Ethical risks (Section 11)
M2 Profile and access control	UserService, RBAC middleware	NFR-11, NFR-29	ER-1, ER-7, ER-14
M3 Scam content analysis	ScamAnalysisService, risk engine	NFR-1, NFR-3	ER-8, ER-9
M4 URL and contact analysis	ScamAnalysisService, URL pipeline	NFR-1, NFR-3	ER-13
M5 Report management	ReportService	NFR-27, NFR-28	ER-1, ER-4, ER-6
M6 Evidence management	EvidenceService, Cloudinary, AuditService	NFR-8, NFR-11	ER-5 , ER-10, ER-12
M7 Review and verification	ReportService, NotificationService, AuditService	NFR-11	ER-10, ER-11
M8 Duplicate detection	ReportService similarity matching	NFR-3	ER-4, ER-13
M9 Community alerts	AlertService	NFR-1	ER-2 , ER-13
M10 Awareness and recovery	Resource and recommendation services	NFR-12, NFR-13	ER-1
M11 Notifications	NotificationService, Nodemailer	NFR-4, NFR-25	ER-6
M12 Admin, reporting and audit	AdministrationService, AnalyticsService, AuditService	NFR-11, NFR-29	ER-3 , ER-7 , ER-10, ER-11

End of Interim Project Report, version 1.0 — Group CyberKent, 1 August 2026.

Version 1.1 will incorporate feedback from the Lecturer, supervisor and Council stakeholders, the resolution of open issues OI-1 to OI-6, and the outcome of the database ethics review at milestone M8.